PyTorch-e GPU Training: Step by Step

CPU theke GPU-te training move korar guide

1. GPU available kina check kora

Shuru korar age dekhte hobe GPU ache kina. Thakle GPU, na thakle CPU use hobe.

```
import torch

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f"Using device: {device}")
```

2. Model-ke GPU-te pathano

Model ke selected device-e (`cuda` ba `cpu`) move korte hobe, jate shob computation ek-i device-e hoy.

```
model = MyModel()          # nijer model diye replace koro
model = model.to(device)   # model GPU-te gelo
```

3. Training loop-e data GPU-te pathano

Proti batch-er features ar labels process korar age GPU-te move korte hobe. Tahole model ar data dui-i ek device-e thake.

```
for batch_features, batch_labels in train_loader:
    # data GPU-te pathao
    batch_features, batch_labels = batch_features.to(device), batch_labels.to(device)
```

4. Evaluation loop-e data GPU-te pathano

Test-er shomoy-o data GPU-te nite hobe. Efficiency-r jonno `torch.no_grad()` diye gradient calculation bondho rakho.

```
with torch.no_grad():
    for batch_features, batch_labels in test_loader:
        # data GPU-te pathao
        batch_features, batch_labels = batch_features.to(device), batch_labels.to(device)
```

5. GPU usage optimize kora

GPU resource-er shobcheye bhalo bebohar korar jonno nicher optimization gulo apply koro.

a. Batch size boro korao

Boro batch size GPU memory-ke bhalo moto kaje lagay ar proti epoch-er computation time komay (jodi memory-te kulay).

```
train_loader = DataLoader(train_dataset, batch_size=128, shuffle=True,  pin_memory=True)
test_loader  = DataLoader(test_dataset,  batch_size=128, shuffle=False, pin_memory=True)
```

b. DataLoader pinning on kora

`DataLoader`-e `pin_memory=True` dile CPU theke GPU-te data transfer druto hoy.

```
train_loader = DataLoader(train_dataset, batch_size=32, shuffle=True, pin_memory=True)
```

Note: original PDF-te 5a-te batch_size=128 ar 5b-te batch_size=32 deya chhilo. Eta shudhu example; 5b-er main point hocche pin_memory=True. Tomar GPU memory onujayi batch size thik

korbe.

Quick summary

Step	Ki korbe	Key code
1	GPU check	`torch.cuda.is_available()`
2	Model move	`model.to(device)`
3	Train data move	`batch.to(device)`
4	Eval data move + no grad	`torch.no_grad()`
5a	Batch size barano	`batch_size=128`
5b	Faster transfer	`pin_memory=True`